Fiscal Year:	FY 2022	**Task Last Updated:**	FY 05/23/2022
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Osteogenic Differentiation of Somatic Stem Cells in Space: A Study Investigating the Role of CDKN1a/p21 on Mesenchymal Stem Cell Proliferation, Differentiation, and Regeneration in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Organization Name:	Rensselaer Polytechnic Institute		
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Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	06/01/2020	**End Date:**	05/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	80NSSC20K1125		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Osteogenic Differentiation of Somatic Stem Cells in Space: A Study Investigating the Role of CDKN1a/p21 on Mesenchymal Stem Cell Proliferation, Differentiation, and Regeneration in Microgravity--ARC" with the same principal investigator, Dr. Elizabeth Blaber, due to PI's move to Rensselaer Polytechnic Institute in summer 2019. Spaceflight factors, including microgravity and space radiation, are known to affect somatic stem cell-based tissue regeneration and cause significant degenerative conditions in multiple mammalian tissues, posing a significant risk for long-term human space exploration. Our previous spaceflight research has shown an inability of both embryonic and somatic stem cells to proliferate, differentiate, and regenerate damaged cells in microgravity conditions, which may be the cause of the widespread tissue degeneration observed in space. Our laboratory has recently published key evidence that spaceflight induces over expression of CDKN1a/p21, a master cell cycle and tissue regeneration inhibitor, in proliferating osteoprogenitor cells, potentially providing a molecular mechanism for spaceflight tissue regenerative deficits. In this proposal, we seek to test the hypothesis that the key molecular mechanism causing the degenerative effects of microgravity on adult stem cells is activation of cell cycle arrest and impairment of differentiation via CDKN1a/p21 overexpression. We specifically hypothesize that in microgravity, normal mouse bone marrow mesenchymal stem cells with the CDKN1a/p21 gene will be inhibited from proliferating and differentiating normally relative to 1g, but that mesenchymal stem cells from CDKN1a/p21-null mice will show unrestrained proliferation and differentiation that is not sensitive to microgravity conditions. To test this hypothesis, we are proposing a spaceflight experiment to Section B.2 ‘Mammalian Cell, Tissue and Organ Generation and Degeneration in Space’ of the NASA Research Announcement NNH14ZTT001N. We specifically propose to conduct an experiment in the NASA Bioculture System Hardware in which the stromal stem cell population isolated from bone marrow of CDKN1a/p21-null and wildtype mice will be taken to the International Space Station (ISS) National Laboratory frozen on the surface of collagen-coated alginate microcarrier beads, injected into bioreactors, and cultured in osteogenic medium to induce differentiation and mineralized bone nodule formation in a 30 day culture. The successful completion of this experiment will conclusively determine the role of CDKN1a/p21 in stem cell regenerative health in space. Specifically, if we observe inhibition of osteogenesis by microgravity in wildtype stem cells, but not in CDKN1a/p21-null cells, this will support our hypothesis that the induction of cell cycle arrest by microgravity in adult stem cells is a key molecular mechanism causing the observed degenerative effects of space on mammalian tissues. This will provide a new perspective on the problem of stem cell regenerative health and medicine in space as well as on Earth, and will provide fundamental insight into stem cell regenerative mechanisms that will aid our understanding of basic stem cell functions on Earth and in space. This research addresses NASA’s and the National Research Council's (NRC’s) Decadal Survey call to study the molecular and cellular mechanisms of bone loss and to conduct basic research into the mechanisms of bone adaptation to microgravity using transgenic animal models. It also will address the call to understand the precise signaling mechanisms responsible for reduced bone formation in space. This research is highly significant, as it proposes to investigate a molecular mechanism, based on previous spaceflight results, that has the potential to explain and mitigate a broad range of microgravity induced tissue degenerative health conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of this experiment will conclusively determine the role of CDKN1a/p21 in stem cell regenerative health in space. Specifically, if we observe inhibition of osteogenesis by microgravity in wildtype stem cells, but not in CDKN1a/p21-null cells, this will support our hypothesis that the induction of cell cycle arrest by microgravity in adult stem cells is a key molecular mechanism causing the observed degenerative effects of space on mammalian tissues. This will provide a new perspective on the problem of stem cell regenerative health and medicine in space as well as on Earth, and will provide fundamental insight into stem cell regenerative mechanisms that will aid our understanding of basic stem cell functions on Earth and in space.
Task Progress:	During this fiscal year we conducted several important experiments to finalise out optimized conditions for spaceflight. These results enable determination of the type of fixative to use in the system that will allow the highest quality and yield of RNA tone extracted upon return, and also enable a thorough characterization of cell thawing protocols and procedures that can be used to initiate cultures in orbit. We also conducted our Science Verification test that enabled the science procedures and parameters we have optimized in a ground based culture system to be translated to the Bioculture System that will be used in orbit. This test was conducted over a 6-week period at NASA Ames Research Center. Unfortunately, due to budgetary constraints and issues with the Bioculture System hardware, Space Biology Management determined that our project will need to be changed to a ground-based study and that the spaceflight portion of the experiment will not continue. Since this determination, we have re-assessed our experimental plan to enable the Aims of the grant to be achieved using samples from the Rodent Research 10 mission and from ground-based studies.
Bibliography Type:	Description: (Last Updated: 04/03/2025)
Abstracts for Journals and Proceedings	Kubik A, Cheng-Campbell MA, Allen NG, Pendyala M, Paul A, Bai Y, Chan D, Blaber EA. "Characterizing the CDKN1a/p21-mediated role of exercise and recovery in mitigating the effect of simulated microgravity on immune dysregulation and systemic tissue degeneration. " 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021